SAT Math Mastery · Session 10

Circles: Equations, Arcs and Tangents

Time	3 hours: Part A in class (about 2 hours of solving), then Part B, the self-assessment, done independently (about 50 minutes), and a short error log.
Topics	circle equations: completing the square, center and radius · writing a circle's equation from conditions · arc length and sector area · chords, diameters and inscribed angles · tangent lines and radii · lines meeting circles · inscribed polygons
Name / Date	_____ _____

PART A · IN CLASS — 35 PROBLEMS

Each problem states its tool. Write your work in the space below. Multiple choice: one correct option. Student-produced response (SPR): enter an exact value (fraction or decimal). Grey boxes show exactly what to type in Desmos, one expression per line; tap a grey dot to read an intersection, intercept or vertex. A letter such as a, c or k becomes a slider when you accept Desmos's offer to add one.

1. HARD · NO CALCULATOR NEEDED

In the xy-plane, the graph of $2x^2 - 8x + 2y^2 + 12y = 24$ is a circle. What is the radius of the circle?

(A) $\sqrt{50}$ **(B) 5 ✓**

(C) 6.5 (D) $\sqrt{40}$

1. Divide every term by 2 so that x^2 and y^2 have coefficient 1: $x^2 - 4x + y^2 + 6y = 12$.
2. Complete the square. Half of -4 is -2 and its square is 4; half of 6 is 3 and its square is 9. Add both squares to each side: the right side is $12 + 4 + 9 = 25$.
3. So $(x-2)^2 + (y+3)^2 = 25$, and the radius is $\sqrt{25} = 5$, choice (B).

Answer: (B)

2. HARD · NO CALCULATOR NEEDED

A circular roundabout has its center at $(2, -5)$ in a coordinate model of the area, where distances are measured in hundreds of meters. Road k is tangent to the roundabout at the point $(9, -1)$. What is the slope of road k?

1. The radius from the center $(2, -5)$ to the point of tangency $(9, -1)$ has slope $\dfrac{-1-(-5)}{9-2} = \dfrac{4}{7}$.
2. A tangent line is perpendicular to the radius at the point of tangency, so its slope is the negative reciprocal: $-\dfrac{7}{4}$.
3. Check: $\dfrac{4}{7} \cdot \left(-\dfrac{7}{4}\right) = -1$, as it should be for perpendicular lines.

Answer: $-\dfrac{7}{4}$

3. HARD · CALCULATOR

A circle has a radius of 12 meters. What is the area, in square meters, of a sector with a central angle of 150°? (Use $\pi \approx 3.14$.)

(A) 150.8 (B) 172.3

(C) 188.4 ✓ (D) 205.6

1. Area of the whole circle: $\pi r^2 = 3.14(12)^2 = 452.16$ square meters.
2. A central angle of 150° is $\frac{150}{360} = \frac{5}{12}$ of the circle.
3. Sector area $= 452.16 \times \frac{5}{12} = 188.4$, which is 188.4 square meters, choice (C).
Answer: (C)

4. HARD · NO CALCULATOR NEEDED

A circle in the xy-plane is defined by the equation $x^2 + y^2 - 6x + 10y + 18 = 0$. What is the distance, in units, from the center of the circle to the origin?

(A) $\sqrt{34}$ ✓ (B) 34

(C) 4 (D) $2\sqrt{34}$

1. Complete the square: $(x^2 - 6x + 9) + (y^2 + 10y + 25) = -18 + 9 + 25$. This gives $(x-3)^2 + (y+5)^2 = 16$.
2. The center is $(3, -5)$ (the signs flip) and the radius is $\sqrt{16} = 4$.
3. Distance from the center to the origin: $\sqrt{3^2 + (-5)^2} = \sqrt{34}$, choice (A). The value 4 is only the radius.
Answer: (A)

5. HARD · NO CALCULATOR NEEDED

A circle in the xy-plane has center $(0, 3)$ and passes through the point $(5, 15)$. Which of the following is an equation of the circle?

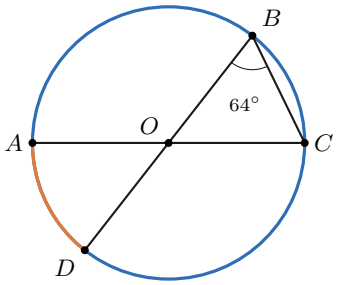
(A) $(x-5)^2 + (y-3)^2 = 169$ (B) $x^2 + (y-3)^2 = 13$

(C) $x^2 + (y-3)^2 = 144$ **(D) $x^2 + (y-3)^2 = 169$ ✓**

1. The radius is the distance from the center $(0, 3)$ to the point $(5, 15)$. We only need $r^2 = (5-0)^2 + (15-3)^2 = 25 + 144 = 169$.
2. Use the center in the equation: $(x-0)^2 + (y-3)^2 = 169$, that is $x^2 + (y-3)^2 = 169$.
3. This is choice (D). Watch the traps: 13 is r (not r^2) and $(x-5)$ would put the point, not the center, in the equation.
Answer: (D)

6. HARD · NO CALCULATOR NEEDED

O is the center of a circle. Chords $\overline{AC}$ and $\overline{BD}$ are diameters that intersect at O (so A, O, C are collinear, and B, O, D are collinear). OB and OC are radii of the circle. If the measure of angle OBC is 64°, what is the measure, in degrees, of arc AD?



1. OB and OC are both radii, so triangle OBC is isosceles and $\angle OCB = \angle OBC = 64°$.
2. The angles of a triangle add to 180°: $\angle BOC = 180 - 64 - 64 = 52°$.
3. AC and BD are diameters crossing at O, so $\angle AOD$ and $\angle BOC$ are vertical angles: $\angle AOD = 52°$.
4. A central angle has the same measure as the arc it cuts off, so arc $AD = 52°$.
Answer: 52

7. HARD · NO CALCULATOR NEEDED

A sector of a circle with radius 9 has an area of 18π. The central angle of the sector measures $k\pi$ radians. What is the value of k?

1. The area of a sector is $\frac{1}{2}r^2\theta$ (with θ in radians): $\frac{1}{2}(9)^2\theta = 18\pi$, that is $\frac{81}{2}\,\theta = 18\pi$.
2. Solve: $\theta = \dfrac{18\pi}{\frac{81}{2}} = \frac{4}{9}\,\pi$.
3. Since $\theta = k\pi$, $k = \frac{4}{9}$ (decimal 0.444 is also accepted).
4. Check: $\frac{4}{9}\pi$ radians is 80°, and $\frac{80}{360} \cdot 81\pi = 18\pi$.
Answer: $\frac{4}{9}$

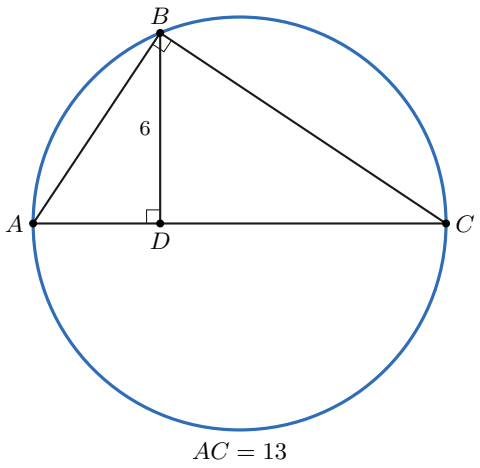
8. HARD · NO CALCULATOR NEEDED

The equation $x^2 + y^2 - 8x + 10y = 8$ defines a circle in the xy-plane. What is the radius of the circle?

1. The equation is $x^2 + y^2 - 8x + 10y = 8$. Half of -8 is -4 (square 16) and half of 10 is 5 (square 25).
2. Add both squares to each side: the right side is $8 + 16 + 25 = 49$.
3. $(x-4)^2 + (y+5)^2 = 49$, so the radius is $\sqrt{49} = 7$.
Answer: 7

9. HARD · NO CALCULATOR NEEDED

Points A, B, and C lie on a circle, and $\angle ABC$ is a right angle. Point D lies on segment AC such that $BD \perp AC$, with $BD = 6$. The diameter of the circle is 13 and $AB < BC$. What is the value of $\dfrac{CD}{AD}$?



1. $\angle ABC = 90^\circ$ is inscribed, so AC is a diameter: $AD + CD = 13$.

2. In right triangle ABC the altitude BD satisfies $BD^2 = AD \cdot CD$, so $AD \cdot CD = 6^2 = 36$.

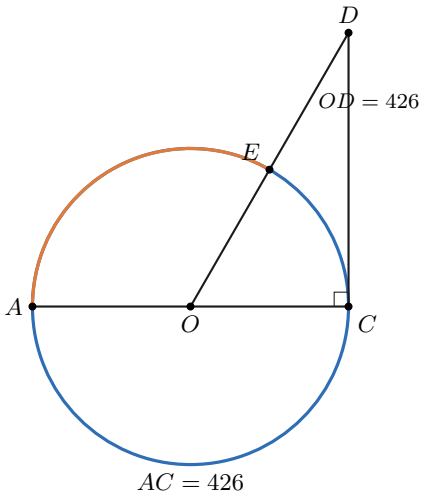
3. AD and CD are the roots of $t^2 - 13t + 36 = 0$, so $t = 4$ or $t = 9$.

4. Since $AB < BC$, point D is nearer to A: $AD = 4$, $CD = 9$, and $\dfrac{CD}{AD} = \dfrac{9}{4}$.

Answer: $\dfrac{9}{4}$

10. HARD · NO CALCULATOR NEEDED

In a circle with center O, segment AC is a diameter and $AC = 426$. Point D lies outside the circle with $OD = AC$, and $\overline{CD}$ is tangent to the circle at C. Segment OD intersects the circle at E. The length of minor arc AE is $b\pi$. What is the value of b?



1. $AC = 426$ is the diameter, so $OC = 213$, and $OD = 426$.
2. The tangent CD is perpendicular to the radius OC, so triangle OCD has a right angle at C and $\cos\angle COD = \dfrac{213}{426} = \dfrac{1}{2}$. So $\angle COD = 60°$.
3. E lies on OD, so $\angle COE = 60°$. A, O, C are in a line, so $\angle AOE = 180° - 60° = 120°$.
4. Arc $AE = \dfrac{120}{360} \cdot 2\pi(213) = 142\pi$, so $b = 142$.

Answer: 142

11. HARD · NO CALCULATOR NEEDED

Circle A in the xy-plane has equation $x^2 + y^2 - 8x + 6y - 8n = 0$, where n is a constant. Circle B has the same center as circle A, has twice the diameter of circle A, and passes through $(10, -3)$. What is the value of n?

1. Completing the square on circle A shows its center is $(4, -3)$ ($x^2 - 8x$ gives $x - 4$ and $y^2 + 6y$ gives $y + 3$).
2. Circle B has the same center and passes through $(10, -3)$, so its radius is $|10 - 4| = 6$. Its diameter is twice that of circle A, so circle A has radius 3.
3. For circle A: $(x - 4)^2 + (y + 3)^2 = 8n + 16 + 9 = 8n + 25$, and this equals $r^2 = 9$.
4. $8n = 9 - 25$, so $n = -2$.

Answer: -2

12. HARD · DESMOS

In the xy-plane, a circle passes through the points $(-1, 2)$, $(7, 2)$ and $(6, 3)$. Which of the following is an equation of the circle?

(A) $(x-3)^2+(y+1)^2=5$

(B) $(x-3)^2+(y-1)^2=17$

(C) $(x-3)^2+(y+1)^2=25$ ✓

(D) $(x-3)^2+(y-2)^2=16$

1. Plot the three points: type `(-1,2)`, `(7,2)` and `(6,3)`.

2. Try the option $(x-3)^2+(y+1)^2=25$: type `(x-3)^2+(y+1)^2=25`. The circle passes through all three points.

3. Check the third point by hand: Type `(6-3)^2+(3+1)^2`: it returns 25, which equals $r^2=25$. The option $(x-3)^2+(y-2)^2=16$ fails: Type `(6-3)^2+(3-2)^2`: it returns 10, not 16, so its circle misses $(6, 3)$.

Answer: (C)

13. HARD · DESMOS

In the xy-plane, the line $x+y=4$ intersects the circle $x^2+y^2-4x+2y=24$ at two points. What is the distance between these two points?

(A) 7

(B) $7\sqrt{2}$ ✓

(C) 14

(D) $\dfrac{7\sqrt{2}}{2}$

1. Type the circle `x^2+y^2-4x+2y=24` and the line `x+y=4`.

2. Tap the two intersection points: (0, 4) and (7, −3).

3. Type `sqrt((7-0)^2+(-3-4)^2)`: it returns 9.8995, which is the distance between them.

4. Exact value: $\sqrt{49+49}=\sqrt{98}=7\sqrt{2}$, and $7\sqrt{2}\approx 9.8995$, choice (B).

Answer: (B)

14. HARD · DESMOS

In the xy-plane, the line $y=k(x-1)-2$, where k is a positive constant, is tangent to the parabola $y=x^2+2x+4$. What is the value of k?

(A) -2

(B) 4

(C) 8

(D) 10 ✓

1. Type the parabola `y=x^2+2x+4`, then the line `y=k(x-1)-2` and accept the slider for k.

2. Move k until the line just touches the parabola at one point. For a positive k this happens at $k=10$; tap the touching point: (4, 28).

3. Check: Type `10*(4-1)-2`: it returns 28 and Type `4^2+2*4+4`: it returns 28, the same height, so the line meets the parabola at $x=4$.

4. The slider also touches at $k=-2$, but the question says k is positive, so $k=10$, choice (D).

Answer: (D)

15. HARD · NO CALCULATOR NEEDED

$$x^2 + y^2 - 6x + 8y = k$$

In the xy-plane, the graph of the equation above, where k is a constant, is a circle that is tangent to the line $4x - 3y = -11$. What is the value of k?

1. Complete the square: $(x-3)^2 + (y+4)^2 = k + 25$ (since $9 + 16 = 25$). The center is $(3, -4)$ and $r^2 = k + 25$.
2. The circle is tangent to the line $4x - 3y + 11 = 0$, so the distance from the center to the line equals the radius: $\dfrac{|4(3) - 3(-4) + 11|}{\sqrt{4^2 + 3^2}} = \dfrac{35}{5} = 7$.
3. So $r = 7$ and $r^2 = 49$. Then $k + 25 = 49$ gives $k = 24$.

Answer: 24

16. HARD · NO CALCULATOR NEEDED

$$(x - a - 2)^2 + (y - 2a + 1)^2 = 25$$

In the xy-plane, the circle above is defined for each real number a. Which of the following points lies on the circle for every real value of a?

(A) $(a + 5,\ 2a + 3)$ ✓

(B) $(a + 5,\ 2a + 4)$

(C) $(2a + 5,\ 2a + 3)$

(D) $(a - 1,\ 2a + 2)$

1. The center is $(a + 2,\ 2a - 1)$ and the radius is 5 (read the signs the other way).
2. A point that is on the circle for every a must always be 5 units from the center, so its offset from the center cannot depend on a. Try the offset $(3, 4)$, since $3^2 + 4^2 = 25$.
3. Center $+(3, 4) = (a + 2 + 3,\ 2a - 1 + 4) = (a + 5,\ 2a + 3)$, choice (A).
4. The others fail: $(a - 1,\ 2a + 2)$ has offset $(-3, 3)$ and $9 + 9 = 18$; $(a + 5,\ 2a + 4)$ has offset $(3, 5)$ and $9 + 25 = 34$; $(2a + 5,\ 2a + 3)$ has offset $(a + 3, 4)$, which depends on a.

Answer: (A)

17. HARD · NO CALCULATOR NEEDED

$$(x-1)^2+(y+3)^2=40$$

$$x^2+y^2-4kx+2ky=9$$

In the xy-plane, the two equations above define circles. The center of the second circle lies on the first circle. If k is a positive constant, what is the value of k?

(A) $1+2\sqrt{7}$ **(B) $1+\sqrt{7}$ ✓**

(C) $-1+\sqrt{7}$ (D) $2+\sqrt{7}$

1. Complete the square in the second circle: x^2-4kx gives $x-2k$ and y^2+2ky gives $y+k$. Its center is $(2k,\,-k)$.
2. That center is on the first circle: $(2k-1)^2+(-k+3)^2=40$, so $4k^2-4k+1+k^2-6k+9=40$, which is $5k^2-10k-30=0$, or $k^2-2k-6=0$.
3. Quadratic formula: $k=\dfrac{2\pm\sqrt{28}}{2}=1\pm\sqrt{7}$ (since $\sqrt{28}=2\sqrt{7}$). The two values are about 3.646 and -1.646.
4. Because $k>0$, $k=1+\sqrt{7}$, choice (B). The value $-1+\sqrt{7}\approx 1.646$ is positive too, but it does not satisfy the equation: $k^2-2k-6\approx -6.58$.

Answer: (B)

18. HARD · NO CALCULATOR NEEDED

In the xy-plane, a circle has equation $x^2+y^2+Dx+Ey+F=0$, where D, E, and F are constants. The circle passes through the points $(2,7)$ and $(10,7)$, has radius 5, and has its center below the line $y=7$. What is the value of $E+F$?

1. The points $(2,7)$ and $(10,7)$ are on the circle and have the same y, so the center lies on the line $x=\dfrac{2+10}{2}=6$. Call it $(6,\,m)$.
2. Radius 5: $(6-2)^2+(m-7)^2=25$, so $(m-7)^2=9$ and $m=7\pm 3$. The center is below $y=7$, so $m=4$.
3. The circle is $(x-6)^2+(y-4)^2=25$, which expands to $x^2+y^2-12x-8y+27=0$ (since $36+16-25=27$).
4. So $E=-8$, $F=27$, and $E+F=19$.

Answer: 19

19. HARD · NO CALCULATOR NEEDED

$$x^2 + y^2 - 4x + 6y - 12 = 0$$

In the xy-plane, the graph of the equation above is a circle. Line ℓ has slope $\frac{3}{4}$ and is tangent to the circle at a point with a positive y-coordinate. What is the y-intercept of ℓ?

(A) $-\frac{43}{4}$ (B) 1

(C) $\frac{7}{4}$ ✓ (D) $-\frac{1}{3}$

1. Complete the square: $(x-2)^2 + (y+3)^2 = 25$. The center is $(2, -3)$ and $r = 5$.
2. A line of slope $\frac{3}{4}$ is $y = \frac{3}{4}x + b$, that is $3x - 4y + 4b = 0$. It is tangent when its distance from the center equals 5: $\frac{|3(2) - 4(-3) + 4b|}{5} = 5$, so $|18 + 4b| = 25$.
3. That gives $4b = 7$ or $4b = -43$, so $b = \frac{7}{4}$ or $b = -\frac{43}{4}$.
4. Find where each line touches the circle: for $b = \frac{7}{4}$ the point is $(-1, 1)$ (positive y); for $b = -\frac{43}{4}$ it is $(5, -7)$ (negative y). So $b = \frac{7}{4}$, choice (C).

Answer: (C)

20. HARD · NO CALCULATOR NEEDED

In the xy-plane, the circle with equation $(x-7)^2 + (y+2)^2 = 9$ has two distinct tangent lines that pass through the point $(2, -2)$. What is the product of the slopes of these two lines?

1. The circle has center $(7, -2)$ and radius 3. The point $(2, -2)$ is on the same horizontal line as the center, 5 units to its left.
2. Each tangent line makes an angle with that horizontal line whose sine is $\frac{r}{d} = \frac{3}{5}$. In this 3-4-5 triangle the tangent length is 4, so the slope has size $\frac{3}{4} = \frac{3}{4}$.
3. The two tangents are mirror images across the horizontal line, so the slopes are $\frac{3}{4}$ and $-\frac{3}{4}$.
4. Product $= \frac{3}{4} \cdot \left(-\frac{3}{4}\right) = -\frac{9}{16}$.

Answer: $-\frac{9}{16}$

21. HARD · DESMOS

In the xy-plane, a circle lies in the first quadrant and is tangent to both the x-axis and the y-axis. The line $3x + 4y = 20$ intersects the circle at two points 8 units apart. What is the sum of all possible values of the radius of the circle?

1. The circle is tangent to both axes in the first quadrant, so its center is (r, r) and its radius is r. The distance from the center to the line $3x + 4y - 20 = 0$ is $\frac{|3r + 4r - 20|}{5} = \frac{|7r - 20|}{5}$.
2. The chord is 8 long, so half of it is 4, and $r^2 = 4^2 + \left(\frac{7r - 20}{5}\right)^2$. Multiply by 25: $25r^2 = 400 + 49r^2 - 280r + 400$, which simplifies to $3r^2 - 35r + 100 = 0$.
3. Type `y=3x^2-35x+100` and tap the two x-intercepts: (5, 0) and (6.6667, 0).
4. Type `5+20/3`: it returns 11.6667, which is $\frac{35}{3}$ (also $-\frac{-35}{3}$ by the sum of roots).

Answer: $\frac{35}{3}$

22. HARD · NO CALCULATOR NEEDED

In the xy-plane, the center of a circle lies on the y-axis. The circle passes through the points $(6, 1)$ and $(-2, 5)$. Which of the following is an equation of the circle?

(A) $x^2 + (y+1)^2 = 40$ ✓

(B) $x^2 + (y-1)^2 = 40$

(C) $x^2 + (y+1)^2 = 20$

(D) $x^2 + (y+1)^2 = 44$

1. The center is $(0, k)$. It is the same distance from $(6, 1)$ and $(-2, 5)$: $36 + (1-k)^2 = 4 + (5-k)^2$.

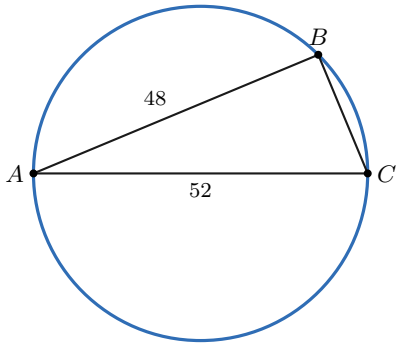
2. Expand: $37 - 2k = 29 - 10k$, so $8k = -8$ and $k = -1$.

3. $r^2 = 6^2 + (1-(-1))^2 = 36 + 4 = 40$, so the circle is $x^2 + (y+1)^2 = 40$, choice (A).

Answer: (A)

23. HARD · NO CALCULATOR NEEDED

Triangle ABC is inscribed in a circle with circumference 52π. Side AC has length 52 and side AB has length 48. What is the ratio of the length of the shortest side of the triangle to the radius of the circle?



(A) $\frac{24}{13}$

(B) $\frac{5}{12}$

(C) $\frac{5}{13}$

(D) $\frac{10}{13}$ ✓

1. Circumference $2\pi r = 52\pi$, so $r = 26$ and the diameter is 52. Since $AC = 52$ equals the diameter, AC is a diameter and $\angle B = 90°$.

2. In right triangle ABC: $BC = \sqrt{52^2 - 48^2} = \sqrt{400} = 20$.

3. The sides are 48, 20, 52, so the shortest is 20.

4. Ratio to the radius: $\frac{20}{26} = \frac{10}{13}$, choice (D).

Answer: (D)

24. HARD · NO CALCULATOR NEEDED

In the xy-plane, the line $y = mx + 10$, where m is a constant, intersects the circle $x^2 + y^2 = 12$ at exactly one point. What is the value of m^2?

1. One common point means the line is tangent to the circle. The circle has center $(0, 0)$ and radius $\sqrt{12}$; the line is $mx - y + 10 = 0$.
2. Distance from the center to the line: $\dfrac{10}{\sqrt{m^2+1}} = \sqrt{12}$. Square both sides: $\dfrac{100}{m^2+1} = 12$.
3. So $m^2 + 1 = \dfrac{100}{12} = \dfrac{25}{3}$, and $m^2 = \dfrac{25}{3} - 1 = \dfrac{22}{3}$.

Answer: $\dfrac{22}{3}$

25. HARD · NO CALCULATOR NEEDED

In the xy-plane, the circle $x^2 + y^2 + 8x - 2y - 8 = 0$ and the line $5x + 12y = 44$ intersect at two points, P and Q. Let C be the center of the circle. What is the area of triangle CPQ?

(A) $\dfrac{25}{2}$ (B) 6

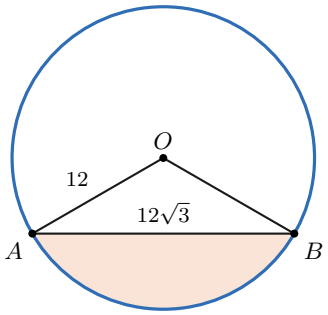
(C) 12 ✓ (D) 24

1. Complete the square: $(x+4)^2 + (y-1)^2 = 25$, so the center is $C(-4, 1)$ and $r = 5$.
2. The distance from C to the line $5x + 12y - 44 = 0$ is $\dfrac{|5(-4) + 12(1) - 44|}{\sqrt{5^2 + 12^2}} = \dfrac{52}{13} = 4$. This is the height of triangle CPQ on base PQ.
3. Half of the chord is $\sqrt{25 - 4^2} = 3$, so $PQ = 6$.
4. Area $= \dfrac{1}{2} \cdot 6 \cdot 4 = 12$, choice (C).

Answer: (C)

26. HARD · NO CALCULATOR NEEDED

In the circle shown, O is the center, the radius is 12, and chord $\overline{AB}$ has length $12\sqrt{3}$. What is the area of the region bounded by chord $\overline{AB}$ and the minor arc AB?



(A) $24\pi - 36\sqrt{3}$

(B) $48\pi - 36\sqrt{3}$ ✓

(C) 48π

(D) $96\pi + 36\sqrt{3}$

1. In triangle OAB the sides are 12, 12, $12\sqrt{3}$. Law of cosines: $\cos\angle AOB = \dfrac{144+144-432}{2(12)(12)} = -\dfrac{1}{2}$, so $\angle AOB = 120°$.

2. Sector area $= \dfrac{120}{360}\pi(12)^2 = 48\pi$.

3. Triangle area $= \dfrac{1}{2}(12)(12)\sin 120° = 36\sqrt{3}$.

4. The region between the chord and the minor arc is sector minus triangle: $48\pi - 36\sqrt{3}$, choice (B).

Answer: (B)

27. HARD · NO CALCULATOR NEEDED

A circular clock face has radius 15 centimeters. A shaded sector of the clock face has area 45π square centimeters. What is the perimeter, in centimeters, of the shaded sector (the arc together with the two bounding radii)?

(A) $15 + 6\pi$

(B) $30 + 3\pi$

(C) $30 + 45\pi$

(D) $30 + 6\pi$ ✓

1. Sector area is $\dfrac{1}{2}r^2\theta$: $\dfrac{1}{2}(15)^2\theta = 45\pi$, so $\theta = \dfrac{45\pi}{\frac{225}{2}} = \dfrac{2}{5}\pi$ radians.

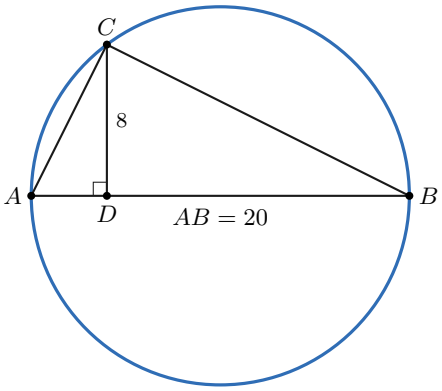
2. Arc length $= r\theta = 15 \cdot \dfrac{2}{5}\pi = 6\pi$.

3. Perimeter = two radii + arc $= 30 + 6\pi$, choice (D). Forgetting the radii gives only the arc.

Answer: (D)

28. HARD · NO CALCULATOR NEEDED

Segment $\overline{AB}$ is a diameter of a circle and has length 20. Point C lies on the circle, and point D lies on $\overline{AB}$ with $\overline{CD} \perp \overline{AB}$ and $CD = 8$. If $AD < DB$, what is the length of $\overline{AC}$?



(A) $4\sqrt{5}$ ✓

(B) $8\sqrt{5}$

(C) $4\sqrt{17}$

(D) 10

1. C is on the circle with diameter AB, so $\angle ACB = 90°$, and CD is the altitude to the hypotenuse: $CD^2 = AD \cdot DB = 64$.
2. With $AD + DB = 20$, AD and DB are the roots of $t^2 - 20t + 64 = 0$: $t = 4$ or $t = 16$. Since $AD < DB$, $AD = 4$.
3. In right triangle ADC: $AC^2 = 4^2 + 8^2 = 80$, so $AC = \sqrt{80} = 4\sqrt{5}$, choice (A).
4. (For comparison, $CB = \sqrt{320} = 8\sqrt{5}$, the longer leg.)
Answer: (A)

29. HARD · NO CALCULATOR NEEDED

A circle has area 48π. A square and an equilateral triangle are each inscribed in the circle (all vertices on the circle). By how much does the area of the square exceed the area of the triangle?

(A) $96 - 27\sqrt{3}$

(B) $96 - 36\sqrt{3}$ ✓

(C) $96 - 12\sqrt{3}$

(D) $192 - 36\sqrt{3}$

1. Area $\pi r^2 = 48\pi$, so $r^2 = 48$.
2. Square: its diagonal is a diameter, $2r$, so area $= \dfrac{(2r)^2}{2} = 2r^2 = 96$.
3. Equilateral triangle in a circle: side $= r\sqrt{3}$, so area $= \dfrac{\sqrt{3}}{4}(r\sqrt{3})^2 = \dfrac{3\sqrt{3}}{4}r^2 = 36\sqrt{3}$.
4. Difference: $96 - 36\sqrt{3}$, choice (B).
Answer: (B)

30. HARD · NO CALCULATOR NEEDED

In the xy-plane, the circle $x^2 + y^2 + 8x - 2y = 8$ and the line $5x - 12y = k$, where k is a constant, intersect at exactly one point. What is the greatest possible value of k?

(A) 65 (B) 32

(C) 33 ✓ (D) -97

1. Complete the square: $(x+4)^2 + (y-1)^2 = 25$, center $(-4, 1)$, $r = 5$.

2. One common point means tangent: the distance from the center to $5x - 12y - k = 0$ equals 5: $\dfrac{|5(-4) - 12(1) - k|}{13} = \dfrac{|k+32|}{13} = 5$.

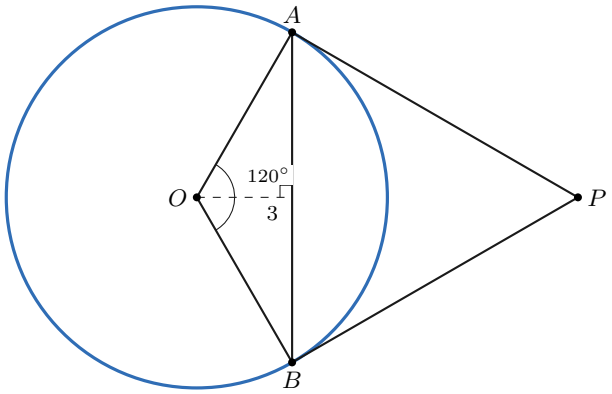
3. So $|k + 32| = 65$, giving $k = 33$ or $k = -97$.

4. The greatest value is 33, choice (C).

Answer: (C)

31. HARD · NO CALCULATOR NEEDED

In a circle with center O, chord $\overline{AB}$ has $\angle AOB = 120°$ and lies at a perpendicular distance of 3 from O. The lines tangent to the circle at A and at B intersect at point P. What is the length of $\overline{PA}$?



(A) $6\sqrt{3}$ ✓ (B) 12

(C) $3\sqrt{3}$ (D) 6

1. The perpendicular from O to chord AB bisects $\angle AOB$, so it makes a $60°$ angle with OA. In that right triangle $\cos 60° = \dfrac{3}{OA}$, so $OA = \dfrac{3}{\frac{1}{2}} = 6$.

2. Tangents at A and B meet at P on the bisector of $\angle AOB$, so $\angle AOP = 60°$, and $PA \perp OA$.

3. In right triangle OAP: $PA = OA \tan 60° = 6\sqrt{3}$, choice (A).

Answer: (A)

32. HARD · NO CALCULATOR NEEDED

In the xy-plane, a circle is tangent to both the x-axis and the y-axis, has its center in the second quadrant, and passes through the point $(-2, 9)$. The diameter of the circle is less than 15. The circle has equation $x^2+y^2+Dx+Ey+F = 0$, where D, E and F are constants. What is the value of F?

1. Tangent to both axes with its center in the second quadrant: center $(-r, r)$ and radius r.
2. It passes through $(-2, 9)$: $(-2 + r)^2 + (9 - r)^2 = r^2$, so $r^2 - 22r + 85 = 0$, that is $(r - 5)(r - 17) = 0$.
3. The diameter must be less than 15: $r = 5$ gives 10, but $r = 17$ gives 34. So $r = 5$.
4. $(x + 5)^2 + (y - 5)^2 = 25$ expands to $x^2 + y^2 + 10x - 10y + 25 = 0$, so $F = 25$.
Answer: 25

33. HARD · NO CALCULATOR NEEDED

$$x^2 + y^2 + 10x - 24y = 0$$

In the xy-plane, the graph of the equation above is a circle. What is the diameter of the circle?

1. Add $(5)^2 = 25$ and $(-12)^2 = 144$ to each side: $(x + 5)^2 + (y - 12)^2 = 25 + 144$.
2. So $r^2 = 169$ and $r = 13$.
3. The question asks for the diameter: $2r = 26$.
Answer: 26

34. HARD · NO CALCULATOR NEEDED

In the xy-plane, a circle has its center on the line $y = 2x$ and passes through the points $(4, 14)$ and $(-1, -1)$. What is the square of the radius of the circle?

1. The center is on $y = 2x$, so write it as $(t, 2t)$.
2. It is equally far from $(4, 14)$ and $(-1, -1)$: $(t-4)^2+(2t-14)^2 = (t+1)^2+(2t+1)^2$, which is $5t^2-64t+212 = 5t^2+6t+2$.
3. The t^2 terms cancel: $210 = 70t$, so $t = 3$ and the center is $(3, 6)$.
4. $r^2 = (3 - 4)^2 + (6 - 14)^2 = 1 + 64 = 65$.
Answer: 65

35. HARD · NO CALCULATOR NEEDED

In the xy-plane, the circle $(x-2)^2+(y+1)^2=25$ passes through the point $(6,2)$. The line tangent to the circle at $(6,2)$ crosses the x-axis at the point $(a,0)$. What is the value of a?

(A) $\dfrac{26}{3}$ (B) $\dfrac{3}{2}$

(C) $\dfrac{10}{3}$ **(D) $\dfrac{15}{2}$ ✓**

1. The radius from the center $(2,-1)$ to $(6,2)$ has slope $\dfrac{2-(-1)}{6-2}=\dfrac{3}{4}$, so the tangent (perpendicular) has slope $-\dfrac{4}{3}$.

2. Tangent line: $y-2=-\dfrac{4}{3}(x-6)$.

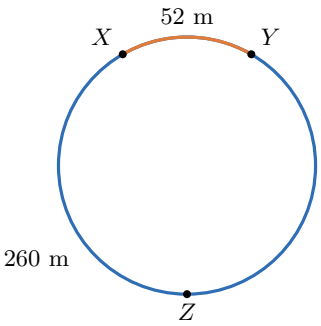
3. Set $y=0$: $-2=-\dfrac{4}{3}(x-6)$, so $x-6=\dfrac{3}{2}$ and $a=\dfrac{15}{2}$, choice (D).

Answer: (D)

PART B · SELF-ASSESSMENT — 21 PROBLEMS

36. MEDIUM · NO CALCULATOR NEEDED

Points X, Y, and Z lie on a circular fountain, as shown below. On this circle, minor arc XY has a length of 52 meters, and major arc XZY has a length of 260 meters. What is the circumference, in meters, of the circle?



(A) 208

(B) 260

(C) 312 ✓

(D) 52

1. The minor arc XY and the major arc XZY together go all the way around the circle, so their lengths add up to the circumference.
2. Circumference $= 52 + 260 = 312$ meters.
3. Sense check: the minor arc is $\frac{1}{6}$ of the circle, which matches a short arc, so choice (C) is right. Choosing 260 alone forgets the minor arc.

Answer: (C)

37. MEDIUM · NO CALCULATOR NEEDED

What is the center of the circle in the xy-plane defined by the equation $x^2 + y^2 - 14x + 22y + 10 = 0$?

(A) $(-7, -11)$

(B) $(7, -11)$ ✓

(C) $(7, 11)$

(D) $(-7, 11)$

1. Group the terms and move the constant: $(x^2 - 14x) + (y^2 + 22y) = -10$.
2. Half of -14 is -7 (square 49); half of 22 is 11 (square 121). The right side becomes $-10 + 49 + 121 = 160$.
3. $(x - 7)^2 + (y + 11)^2 = 160$. The signs flip, so the center is $(7, -11)$, choice (B).

Answer: (B)

38. Medium · No calculator needed

A circle in the xy-plane has equation $(x+2)^2+(y-5)^2=49$. What is the radius of the circle?

(A) 14 (B) 24.5

(C) 49 **(D) 7 ✓**

1. Compare with the standard form $(x-h)^2+(y-k)^2=r^2$: the right side is r^2, not r.
2. Here $r^2=49$, so $r=\sqrt{49}=7$. (The center $(-2,5)$ is not needed.)
3. Choice (D) is the radius; 49 is r^2 and 14 is the diameter.
Answer: (D)

39. Medium · Calculator

A circle has a radius of 9 cm. What is the length, in cm, of an arc that subtends a central angle of 80°? (Use $\pi \approx 3.14$.)

(A) 12.6 ✓ (B) 15.7

(C) 18.8 (D) 9.4

1. Circumference: $2\pi r = 2(3.14)(9) = 56.52$ cm.
2. An angle of 80° is $\dfrac{80}{360}=\dfrac{2}{9}$ of the full circle.
3. Arc length $= 56.52 \times \dfrac{2}{9} = 12.56$, which rounds to 12.6 cm, choice (A).
Answer: (A)

40. Medium · No calculator needed

The equation of a circle in the xy-plane is $x\,(x-8)+y\,(y+6)=39$. What is the radius of the circle?

(A) 7 **(B) 8 ✓**

(C) 9 (D) 6

1. Distribute first: $x(x-8)+y(y+6)=39$ becomes $x^2-8x+y^2+6y=39$.
2. Complete the square: add $(-4)^2=16$ and $3^2=9$ to both sides, so the right side is $39+16+9=64$.
3. $(x-4)^2+(y+3)^2=64$, so $r=\sqrt{64}=8$, choice (B).
Answer: (B)

41. Medium · No calculator needed

A circle has an area of 144π ft^2. What is the circumference of the circle?

(A) 12 ft (B) 12π ft

(C) 24π ft ✓ (D) 48π ft

1. Area $= \pi r^2 = 144\pi$, so $r^2 = 144$ and $r = 12$ ft.
2. Circumference $= 2\pi r = 2\pi(12) = 24\pi$ ft.
3. That is choice (C). The value 12 is the radius; the circumference needs the factor 2π.
Answer: (C)

42. Medium · No calculator needed

A circle has radius $2x - 3$ and circumference $(3x + 4)\pi$. What is the area of the circle?

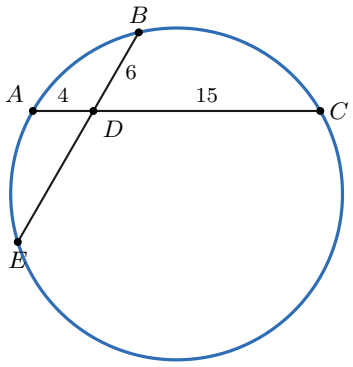
(A) 289π ✓ (B) 17π

(C) 34π (D) 196π

1. Circumference $= 2\pi r$, so $2\pi(2x - 3) = (3x + 4)\pi$. Divide by π: $4x - 6 = 3x + 4$.
2. Solve: $x = 10$.
3. The radius is $2(10) - 3 = 17$. (Check: circumference $34\pi = (3 \cdot 10 + 4)\pi$.)
4. Area $= \pi r^2 = \pi(17)^2 = 289\pi$, choice (A).
Answer: (A)

43. Medium · No calculator needed

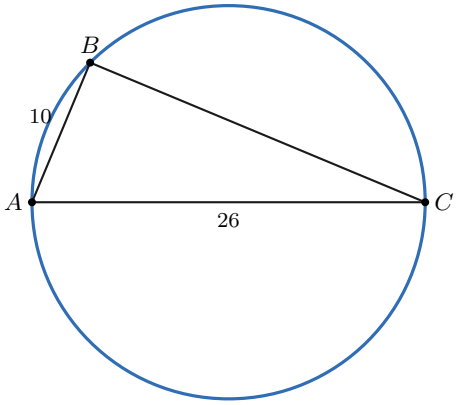
Chords $\overline{AC}$ and $\overline{BE}$ of a circle intersect at point D inside the circle. If $AD = 4$, $DC = 15$, and $BD = 6$, what is the length of $\overline{BE}$?



1. Two chords crossing inside a circle: (piece)(piece) is the same for both, so $AD \cdot DC = BD \cdot DE$.
2. $4 \cdot 15 = 6 \cdot DE$, so $60 = 6 \cdot DE$ and $DE = 10$.
3. $BE = BD + DE = 6 + 10 = 16$.
Answer: 16

44. Medium · No calculator needed

Segment AC is a diameter of a circle, and point B lies on the circle. If $AC = 26$ and $AB = 10$, what is the length of BC?



(A) $\sqrt{776}$

(B) 26

(C) 16

(D) 24 ✓

1. AC is a diameter, so an angle inscribed in the semicircle is a right angle: $\angle ABC = 90°$ and AC is the hypotenuse.
2. Pythagorean theorem: $BC^2 = 26^2 - 10^2 = 676 - 100 = 576$.
3. $BC = \sqrt{576} = 24$, choice (D). (Do not subtract $26 - 10$.)
Answer: (D)

45. Medium · No calculator needed

The equation $x^2 - 18x + y^2 - 6y = -38$ represents a circle in the xy-plane. What is the area of the circle?

(A) $2\sqrt{13}\,\pi$

(B) 43π

(C) 52π ✓

(D) 128π

1. Half of -18 is -9 (square 81) and half of -6 is -3 (square 9). Add both to each side.
2. The right side is $-38 + 81 + 9 = 52$, so $(x-9)^2 + (y-3)^2 = 52$ and $r^2 = 52$.
3. Area $= \pi r^2 = 52\pi$, choice (C). Adding 38 instead of -38 would give 128π.
Answer: (C)

46. Medium · No calculator needed

A circle has radius 15. An arc of the circle has length 6π. What is the measure, in degrees, of the central angle that subtends the arc?

(A) 18 **(B) 72 ✓**

(C) 36 (D) 144

1. Circumference $= 2\pi(15) = 30\pi$.

2. The arc 6π is $\dfrac{6\pi}{30\pi} = \dfrac{1}{5}$ of the whole circle.

3. Central angle $= \dfrac{1}{5} \cdot 360^\circ = 72^\circ$, choice (B).

Answer: (B)

47. Medium · No calculator needed

A sector of a circle with radius 6 has area 12π. What is the measure, in degrees, of the central angle of the sector?

1. The full circle has area $\pi(6)^2 = 36\pi$.

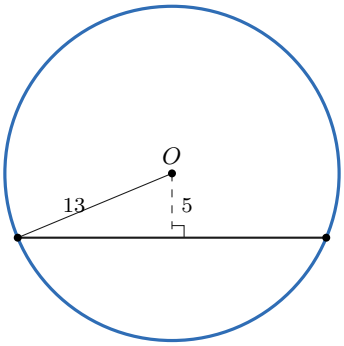
2. The sector is $\dfrac{12\pi}{36\pi} = \dfrac{1}{3}$ of the circle.

3. Central angle $= \dfrac{1}{3} \cdot 360^\circ = 120^\circ$.

Answer: 120

48. Medium · No calculator needed

A chord of a circle with radius 13 is 5 units from the center of the circle. What is the length of the chord?



1. The perpendicular from the center to a chord bisects the chord, giving a right triangle with hypotenuse 13 and one leg 5.

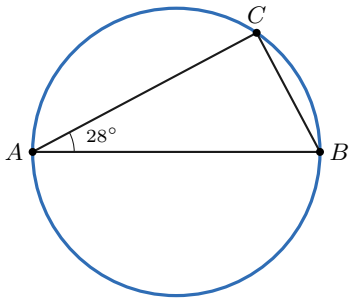
2. The other leg is $\sqrt{13^2 - 5^2} = \sqrt{144} = 12$ (half of the chord).

3. The chord is $2(12) = 24$.

Answer: 24

49. MEDIUM · NO CALCULATOR NEEDED

Points A and B are the endpoints of a diameter of a circle, and point C is on the circle. The measure of angle CAB is 28°. What is the measure, in degrees, of angle ABC?



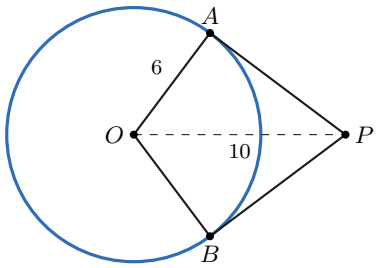
1. AB is a diameter, so an angle inscribed in the semicircle is a right angle: $\angle ACB = 90°$.

2. The angles of triangle ABC add to 180°: $\angle ABC = 180 - 90 - 28 = 62°$.

Answer: 62

50. HARD · NO CALCULATOR NEEDED

Point P is 10 units from the center O of a circle with radius 6. Segments $\overline{PA}$ and $\overline{PB}$ are tangent to the circle at A and B. What is the area of quadrilateral $PAOB$?



1. A tangent is perpendicular to the radius at the point of tangency, so $\angle PAO = 90°$ and $PA = \sqrt{10^2 - 6^2} = \sqrt{64} = 8$.

2. Triangle PAO has area $\frac{1}{2}(8)(6) = 24$.

3. Triangle PBO is congruent to it, so quadrilateral $PAOB$ has area $2(24) = 48$.

Answer: 48

51. Medium · No calculator needed

In the xy-plane, a circle has center $(-3, 5)$ and is tangent to the x-axis. Which of the following is an equation of the circle?

(A) $(x+3)^2 + (y-5)^2 = 9$

(B) $(x+3)^2 + (y-5)^2 = 5$

(C) $(x-3)^2 + (y+5)^2 = 25$

(D) $(x+3)^2 + (y-5)^2 = 25$ ✓

1. A circle tangent to the x-axis touches it directly below or above the center, so the radius equals the center's distance from the axis: $|5| = 5$.

2. $r^2 = 25$, and the center $(-3, 5)$ gives $(x+3)^2 + (y-5)^2 = 25$, choice (D).

Answer: (D)

52. Medium · No calculator needed

In the xy-plane, the line $y = 6$ intersects the circle $x^2 + y^2 = 100$ at two points. What is the distance between the two points?

1. Substitute $y = 6$ into $x^2 + y^2 = 100$: $x^2 + 36 = 100$, so $x^2 = 64$ and $x = \pm 8$.

2. The points are $(-8, 6)$ and $(8, 6)$, which are 16 units apart.

Answer: 16

53. Medium · No calculator needed

An arc of a circle has length 10π centimeters and subtends a central angle of $\dfrac{2\pi}{3}$ radians. What is the radius of the circle, in centimeters?

1. Arc length is $s = r\theta$ with θ in radians: $10\pi = r \cdot \dfrac{2\pi}{3}$.

2. $r = 10\pi \cdot \dfrac{3}{2\pi} = 15$ centimeters.

Answer: 15

54. Medium · No calculator needed

Two concentric circles have the same center. The outer circle has radius 7, and the area of the region between the two circles is 48π. What is the radius of the inner circle?

1. Ring area = outer area − inner area: $\pi(7)^2 - \pi r^2 = 48\pi$.

2. $49 - r^2 = 48$, so $r^2 = 1$.

3. The inner radius is $r = 1$.

Answer: 1

55. Medium · No calculator needed

A sector of a circle has radius 12 and perimeter $24 + 4\pi$. What is the measure, in degrees, of the central angle of the sector?

(A) 60 ✓ (B) 120

(C) 240 (D) 30

1. The perimeter is two radii plus the arc: $2(12) + \text{arc} = 24 + 4\pi$, so the arc is 4π.

2. The arc is $\frac{\theta}{360}$ of the circumference $2\pi(12) = 24\pi$: $\frac{\theta}{360} \cdot 24\pi = 4\pi$.

3. $\theta = 360 \cdot \frac{4}{24} = 60°$, choice (A).

Answer: (A)

56. Hard · No calculator needed

A circle has area 50π. A square is inscribed in the circle so that all four vertices lie on the circle. What is the area of the square?

1. Area $\pi r^2 = 50\pi$, so $r^2 = 50$ and the diameter is $2\sqrt{50}$.

2. The square's diagonal is the diameter. For a square, area $= \frac{(\text{diagonal})^2}{2} = \frac{4(50)}{2} = 100$.

3. So the area of the square is 100 (side 10).

Answer: 100